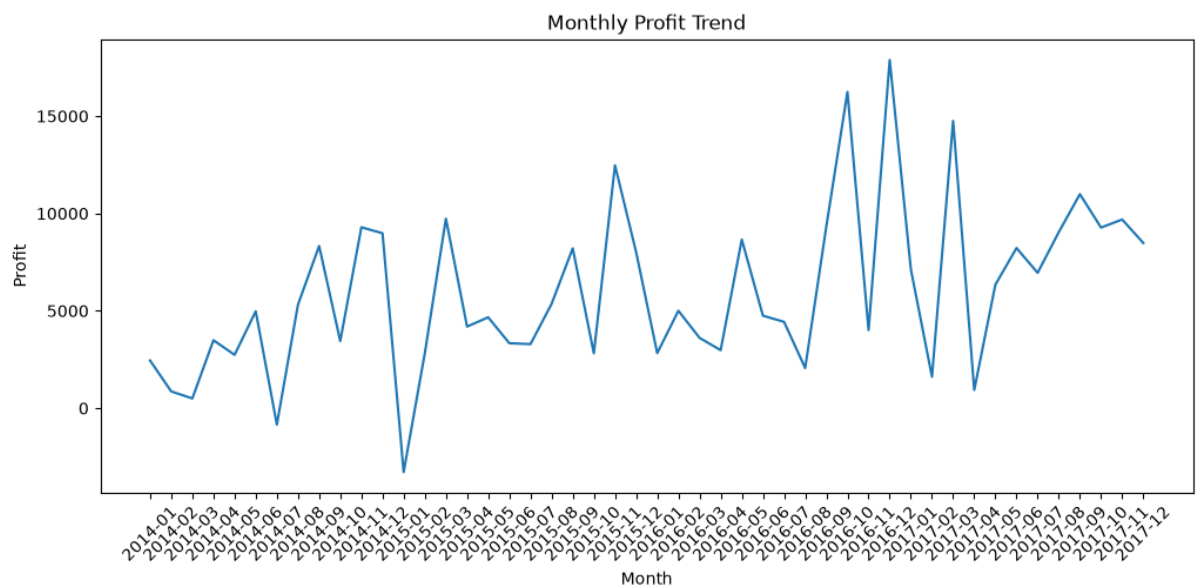
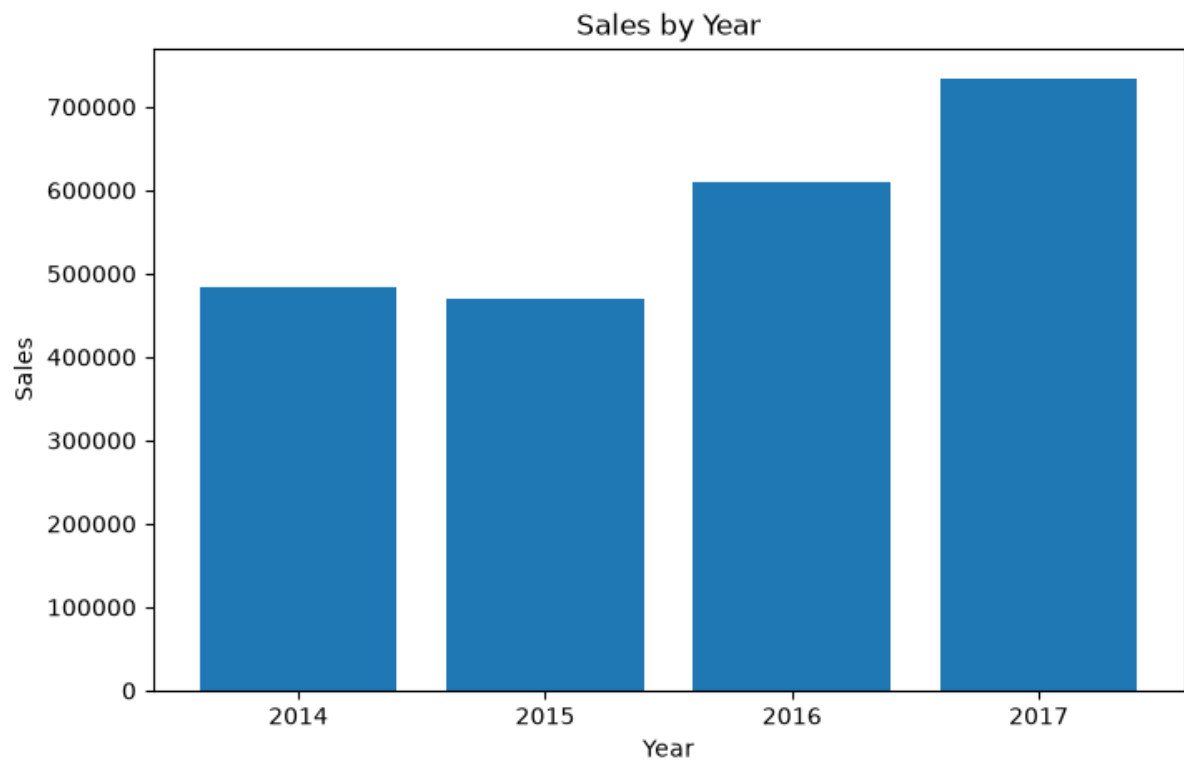
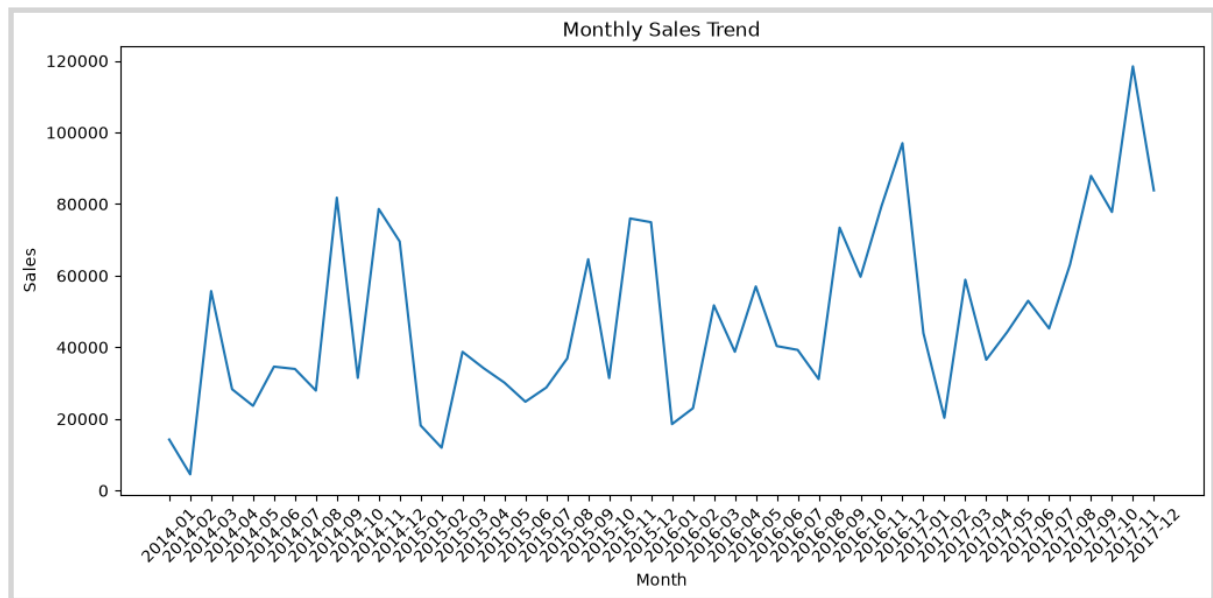


Week 2

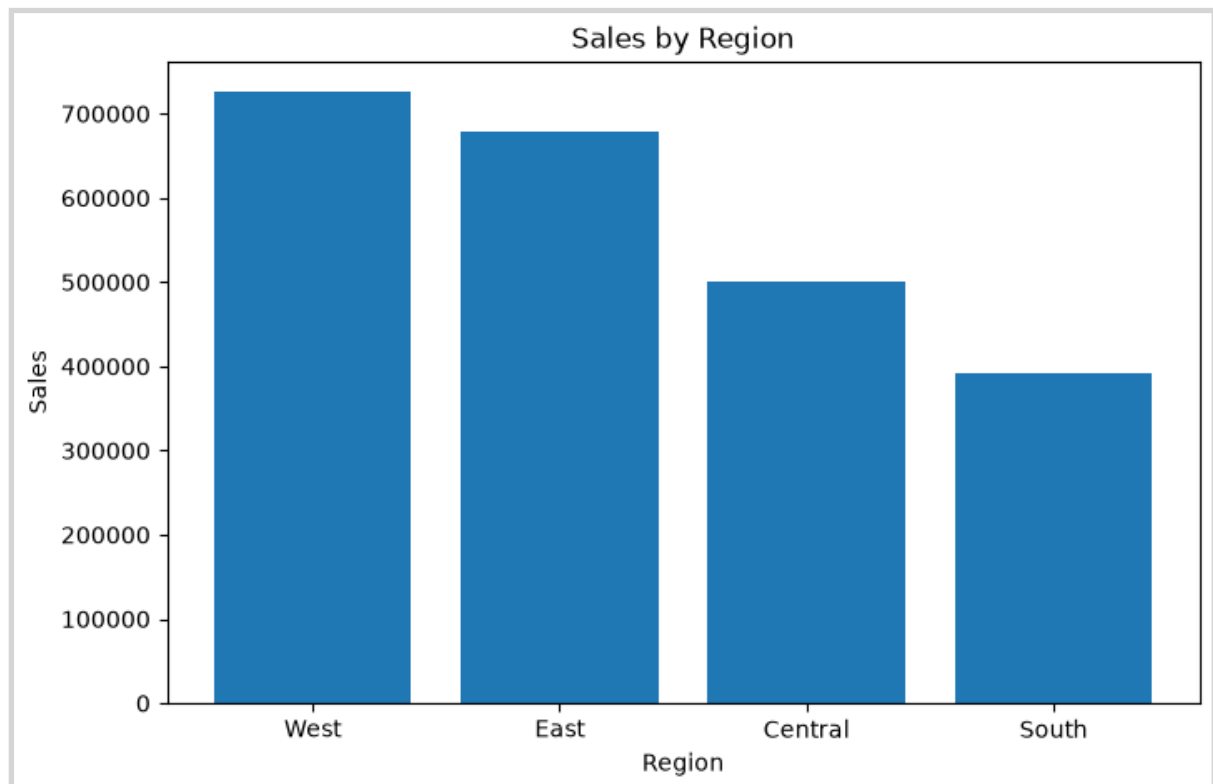
Exploratory Data Analysis & Business Insights

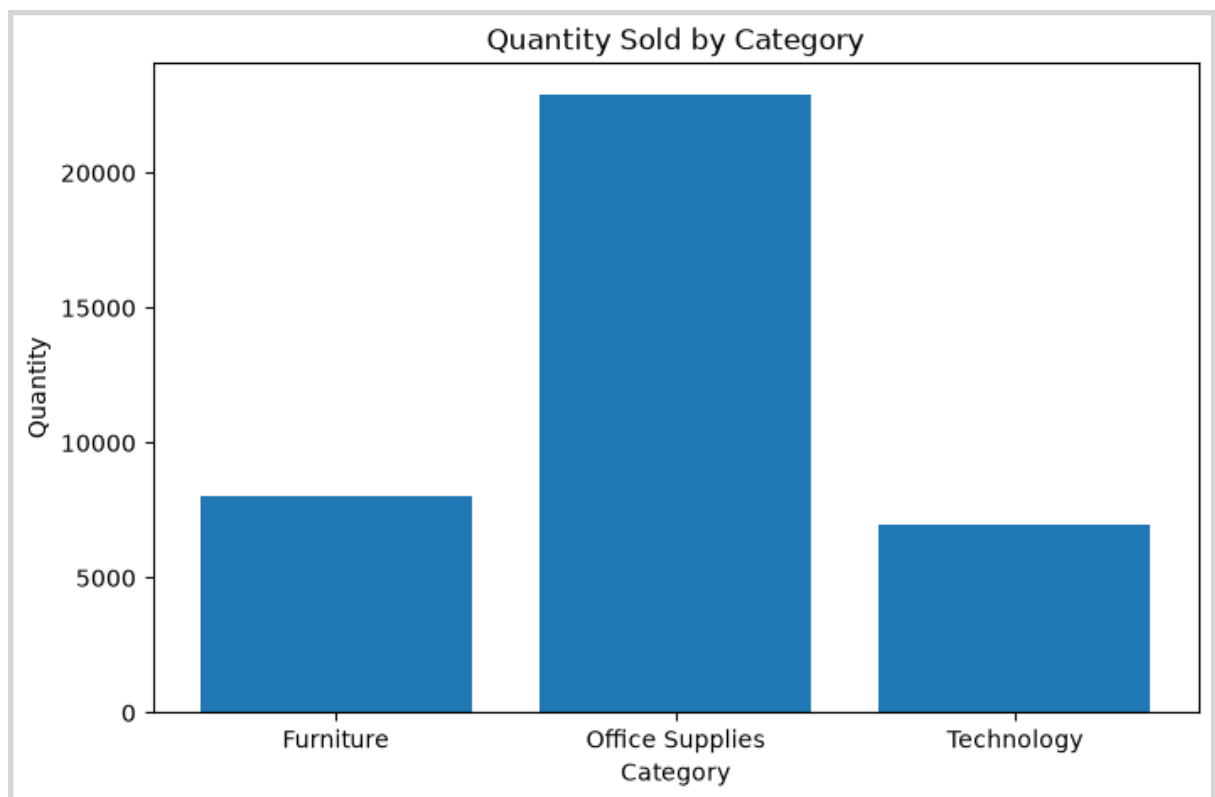
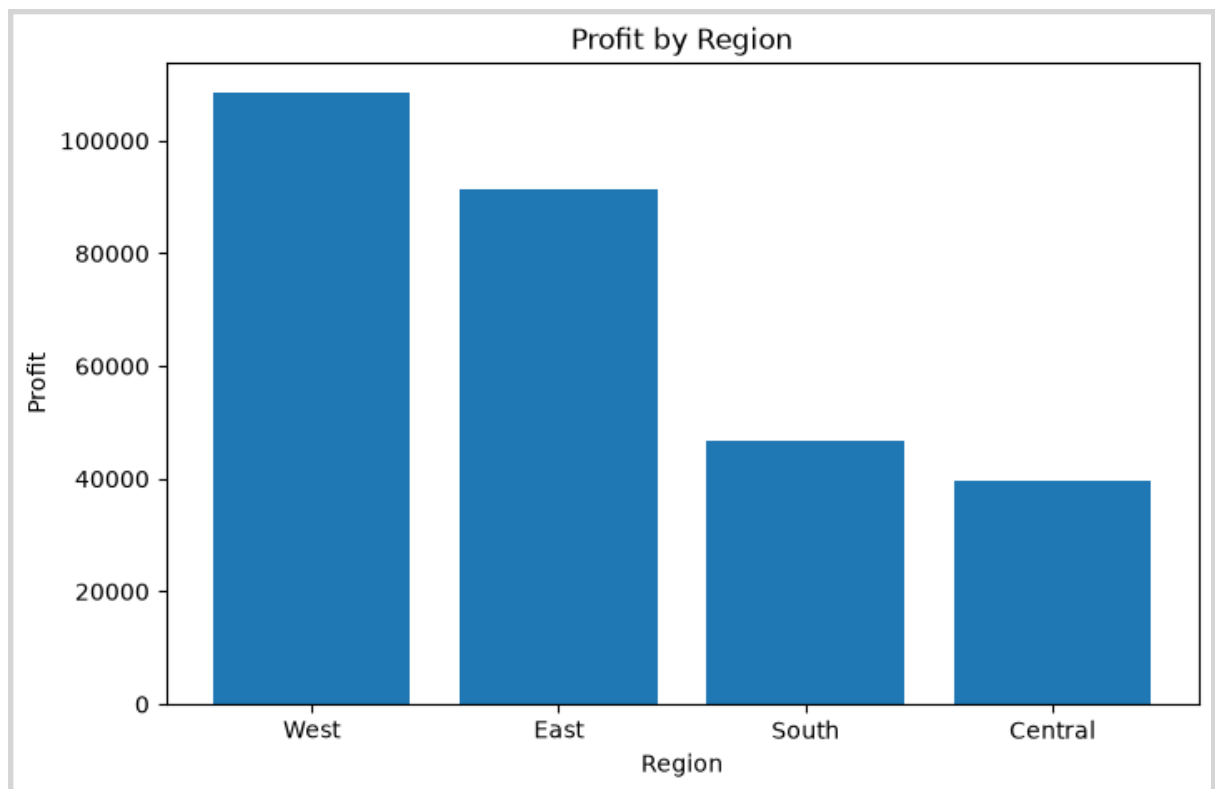
Sales Trend Analysis

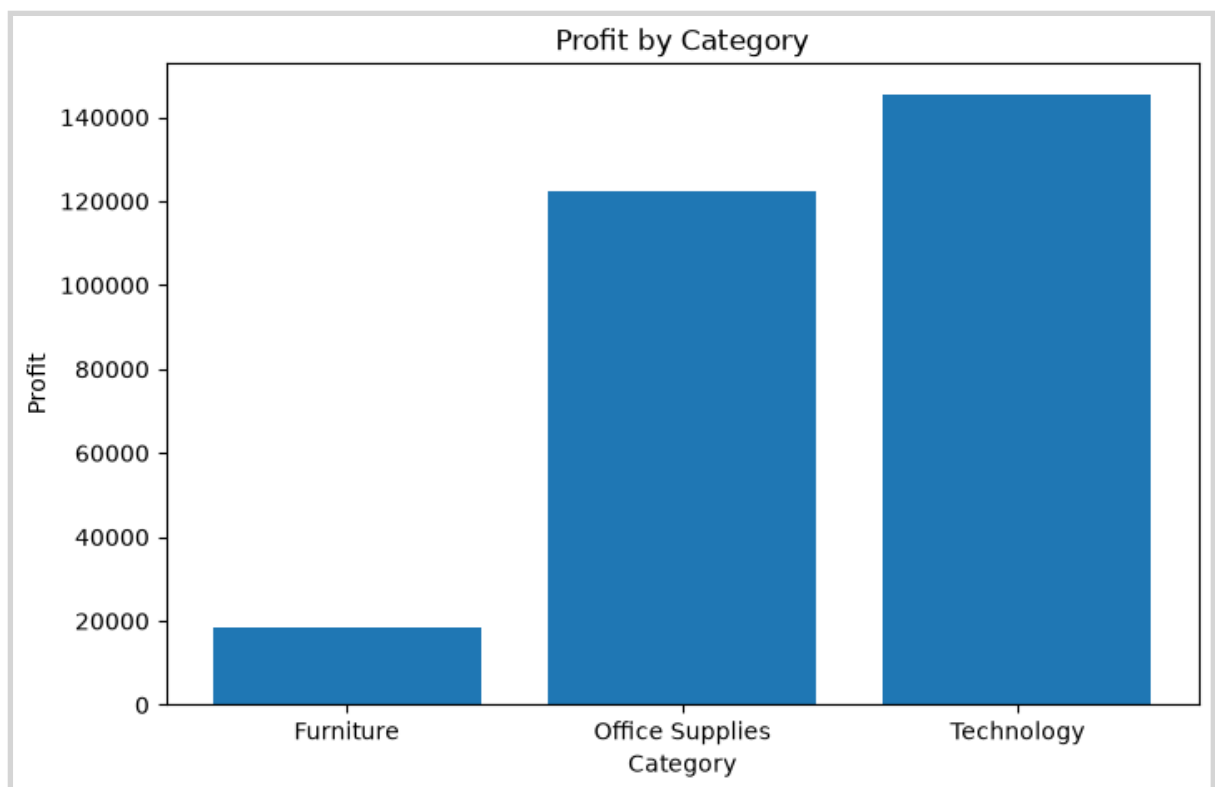
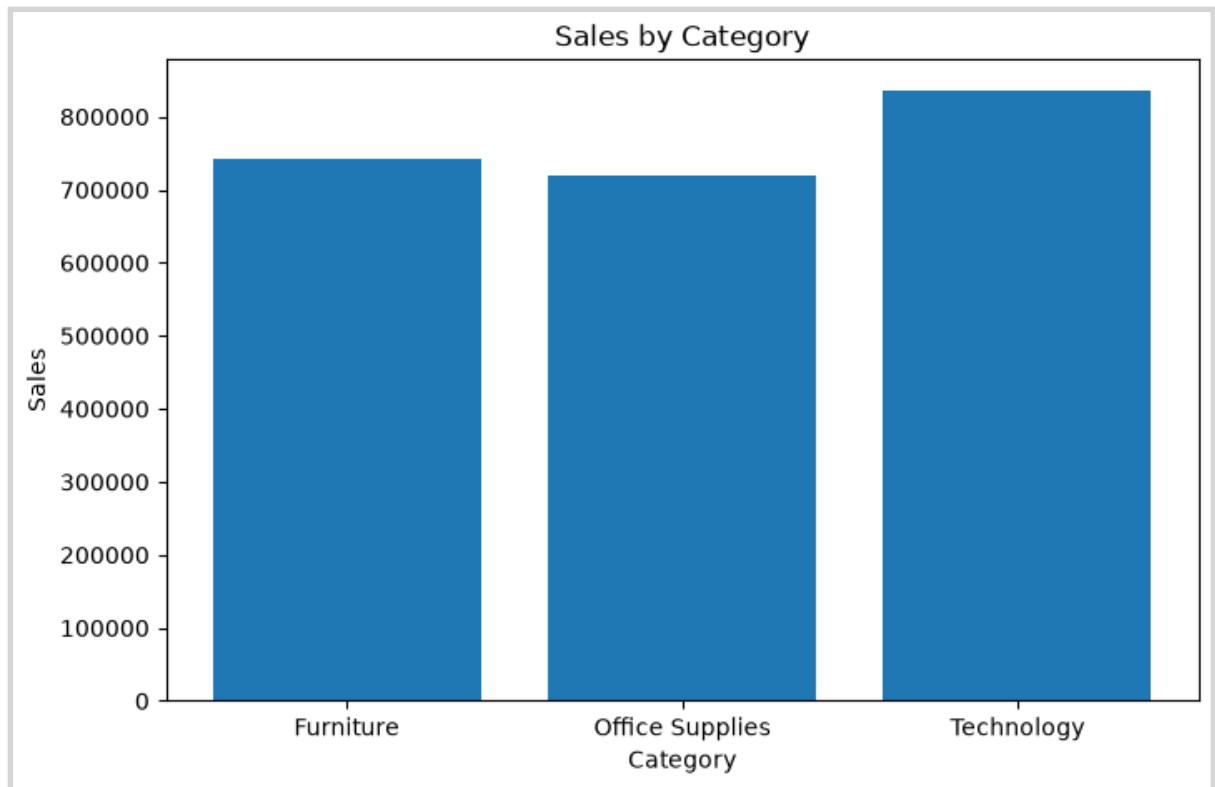




Regional Analysis







Key Business Insights

Based on the exploratory analysis performed in this notebook, the following business insights were identified:

Insight 1 — Overall Performance

The business generated \$2,297,200.86 in total sales** and \$286,397.02 in total profit from 5,009 orders. The overall profit margin was 12.47%, indicating that the business generated a positive return from its recorded sales activity.

Insight 2 — Sales Trend

Monthly sales fluctuated significantly throughout the analysis period, with several noticeable peaks and declines. Overall, sales activity increased toward the later years, with the highest monthly sales occurring in October 2017 at approximately \$118,000.

Insight 3 — Regional Performance

The West region generated the highest sales, with approximately \$725,000 in total sales, followed by the East region with approximately \$680,000. The South region recorded the lowest sales, at approximately \$390,000, indicating a substantial difference in sales contribution across regions.

Insight 4 — Category Performance

Technology was the strongest-performing category, generating approximately \$836,000 in sales, followed by Furniture at approximately \$743,000 and Office Supplies at approximately \$719,000. However, sales volume did not translate evenly into profit — Office Supplies recorded the highest quantity sold at roughly 23,000 units, compared to only about 8,000 units for Furniture and

7,000 for Technology, indicating that Office Supplies operates on a high-volume, lower-price-point model relative to the other two categories.

Insight 5 — Profitability

Profitability varied sharply across categories despite relatively similar sales figures. Technology delivered the highest profit at approximately \$145,500 (a margin of roughly fulfillment expenses relative to its revenue, making it a priority area for margin investigation

17.4%), and Office Supplies followed closely with approximately \$122,000 in profit (a margin of roughly 17%). Furniture, by contrast, generated only about \$18,500 in profit despite sales nearly on par with Technology, resulting in a margin of just 2.5% — the weakest of the three categories. This gap suggests that Furniture's profitability is disproportionately affected by higher costs, discounting, or fulfillment expenses relative to its revenue, making it a priority area for margin investigation